

Practice Question 1

Q1: Consider the following sequence:

A user starts Application A. The executable is stored on disk. The OS creates a process for A. A is scheduled on the CPU and starts running. While running, A requests data from a file. The OS services the request, and another ready process B gets the CPU.

Explain the OS concepts involved in this sequence.

Practice Question 1

- Program → Process
- Program loading / virtual address space
- System call / user mode → kernel mode
- Process state change and scheduling

Expected Answer

The executable initially exists as a program on disk. When the user runs it, the OS creates a process, including a PID, process state, memory image and execution context. The OS creates its virtual address space and maps the program's code and data into it.

The process starts in a state where it can be scheduled and eventually becomes Running. When Application A requests file data, it uses a system call, causing execution to enter kernel mode so the OS can perform the requested service.

If A must wait for the file I/O, A moves from Running → Blocked/Waiting. The scheduler can then select B, causing B to move from Ready → Running. When A's I/O completes, it can move back to Ready.

These concepts are presented across the slides on program loading, system calls, processes and process states.

Practice Question 2

Q2: A company wants to run three applications on one physical server.

Approach A: Each application runs with its own Guest OS on top of a hypervisor.

Approach B: Each application runs in a separate container while sharing the host OS kernel.

Explain the main architectural difference between A and B. Which approach provides full machine/OS virtualization, and which approach primarily provides process/environment isolation?

Practice Question 2

Expected Answer

Approach A represents virtual machines. Each VM has a Guest OS and runs above a hypervisor. The machine and its OS are virtualized.

Approach B represents containers. Containers run isolated application environments while sharing the host OS kernel. The slides describe containers as isolated environments using namespaces, while VMs virtualize the machine/OS layer.

Therefore:

A → Virtualization / VM

B → Containerization / isolation

Practice Question 3

Q3: Suppose Process P1 is running and the OS decides to stop it temporarily so that another process can run.

Explain what information the OS needs to preserve so that P1 can continue correctly later. Also explain why the entire process does not need to be copied during a context switch.

Practice Question 3

Expected Answer

The OS needs to preserve P1's CPU execution context, including the Program Counter, general-purpose registers, Stack Pointer, and processor status/flags. This information allows the CPU to resume P1 from the correct instruction with the correct register and stack state.

The OS does not need to copy the entire process because the process's code, data, heap and stack generally remain in memory. The main operation is saving P1's CPU state and restoring the state of the next process.

Practice Question 4

Q4: A process is executing in user mode and performs a system call. Later, the scheduler switches the CPU from Process P1 to Process P2.

Explain the difference between these two switches:

User mode → Kernel mode

P1 → P2

Why is the second operation generally more expensive?

Practice Question 4

Expected Answer

A mode switch changes the CPU's privilege level from user mode to kernel mode so that the kernel can execute a privileged service. The same process is still executing; only its execution mode changes.

A context switch changes the process/thread executing on the CPU. The OS must save the execution context of P1 and restore the saved context of P2. This can involve the program counter, stack pointer, general-purpose registers and, for a process switch, address-space-related state.

Therefore, a context switch generally involves more state changes than a simple mode switch.

Practice: CPU Scheduling

1. Calculation of Waiting time
2. Calculation of Turnaround time
3. Preemptive and Non-preemptive policies
4. Execution time slots
5. FCFS
6. SJF
7. SRTF
8. RR

Practice

Which combination correctly ranks scheduling algorithms from most preemptive-friendly to strictly non-preemptive: SRTF, Round Robin, FCFS?

1. FCFS, Round Robin, SRTF
2. SRTF and Round Robin are preemptive; FCFS is strictly non-preemptive
3. All three are strictly non-preemptive
4. All three are strictly preemptive

Practice: FCFS

Two processes P1 and P2 arrive at time 0 with burst times 5 ms and 9 ms respectively. Using FCFS scheduling (P1 before P2), compute the waiting time and turnaround time for each, and the average waiting time.

Given:

Process	Arrival	Burst
P1	0	5 ms
P2	0	9 ms

Practice

Process	Arrival	Burst	Completion	Waiting Time	Turnaround Time
P1	0	5	5	0	5
P2	0	9	14	5	14

Average WT = 2.5 ms

Practice: SJF

Four processes P1, P2, P3, P4 arrive at time 0 with burst times 6, 2, 5, 3 ms. Using SJF (non-preemptive) scheduling: (a) Compute the waiting time and turnaround time for each process. (b) Compute the average waiting time and average turnaround time.

Given:

Process	Arrival	Burst
P1	0	6 ms
P2	0	2 ms
P3	0	5 ms
P4	0	3 ms

Practice: SJF

Process	Arrival	Burst	Start	Completion	WT	TAT
P2	0	2	0	2	0	2
P4	0	3	2	5	2	5
P3	0	5	5	10	5	10
P1	0	6	10	16	10	16

Average WT = 4.25 ms

Average TAT = 8.25 ms

Practice: SRTF

Three processes P1, P2, P3 arrive at times 0, 1, and 3 ms with burst times 9, 5, and 2 ms respectively. Using preemptive SRTF scheduling: (a) Draw the Gantt chart. (b) Compute waiting and turnaround time for each process. (c) Compute the average waiting time.

Given:

Process	Arrival	Burst
P1	0	9 ms
P2	1	5 ms
P3	3	2 ms

Practice: SRTF

0	1	3	5	8	16
P1	P2	P3	P2	P1	

Process	Completion	Arrival	TAT
P1	16	0	16
P2	8	1	7
P3	5	3	2

Process	TAT	Burst	WT
P1	16	9	7
P2	7	5	2
P3	2	2	0

Summary

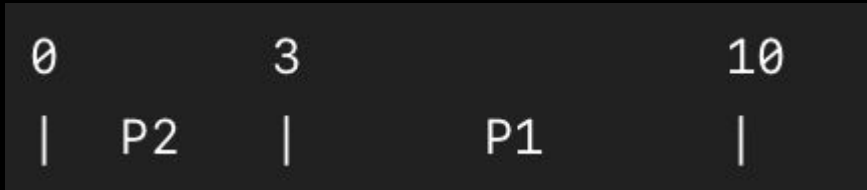
- $TAT = Completion - Arrival$
 - $WT = TAT - Burst$, Therefore:
- $WT = Completion - Arrival - Burst$

Practice: SJF

Two processes P1 and P2 arrive at time 0 with CPU burst times 7 ms and 3 ms. Using SJF (non-preemptive) scheduling, draw the execution order and compute the average turnaround time.

Practice: SJF

Two processes P1 and P2 arrive at time 0 with CPU burst times 7 ms and 3 ms. Using SJF (non-preemptive) scheduling, draw the execution order and compute the average turnaround time.



Process	Waiting Time	Turnaround Time
P1	3 ms	10 ms
P2	0 ms	3 ms

Practice: FCFS

Four processes P1, P2, P3, P4 have burst times 4, 7, 2, 5 ms and all arrive at time 0. Using FCFS scheduling algorithm (a) Compute waiting time and turnaround time for each. (b) Compute the average waiting time and average turnaround time.

Practice: FCFS

Four processes P1, P2, P3, P4 have burst times 4, 7, 2, 5 ms and all arrive at time 0. Using FCFS scheduling algorithm (a) Compute waiting time and turnaround time for each. (b) Compute the average waiting time and average turnaround time.

Process	Waiting Time	Turnaround Time
P1	0 ms	4 ms
P2	4 ms	11 ms
P3	11 ms	13 ms
P4	13 ms	18 ms

Average Waiting Time: 7ms

Average Turnaround Time: 11.5ms

Practice: SRTF

Three processes P1, P2, P3 arrive at times 0, 2, and 4 ms with CPU burst times 7, 4, and 1 ms respectively. Using preemptive SRTF scheduling: (a) Compute waiting and turnaround time for each process. (b) Compute the average turnaround time.

Practice: SRTF

Three processes P1, P2, P3 arrive at times 0, 2, and 4 ms with CPU burst times 7, 4, and 1 ms respectively. Using preemptive SRTF scheduling: (a) Compute waiting and turnaround time for each process. (b) Compute the average turnaround time.

Process	Completion	Waiting Time	Turnaround Time
P1	12 ms	5 ms	12 ms
P2	7 ms	1 ms	5 ms
P3	5 ms	0 ms	1 ms

Average Turnaround Time: 6ms

Practice Assignment

1. Three processes P1, P2, P3 arrive at time 0 with CPU burst times of 6, 2, and 4 ms respectively. Using the SJF (non-preemptive) scheduling algorithm, draw the execution order and compute the average waiting time.
2. Four processes P1, P2, P3, P4 arrive at time 0 with CPU burst times 5, 3, 8, 6 ms respectively. Using FCFS scheduling algorithm (a) Compute the waiting time and turnaround time for each process. (b) Compute the average waiting time and average turnaround time.
3. Three processes P1, P2, P3 arrive at times 0, 1, and 2 ms with CPU burst times 8, 4, and 9 ms respectively. Using preemptive SRTF (Shortest Remaining Time First) scheduling: (a) Compute the waiting time and turnaround time for each process. (b) Compute the average waiting time.